Research Proposal: Active Learning for Long-term Robotics

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1 Abstract

Recent advances in vision–language–action (VLA) models have significantly improved robotic manipulation in unstructured environments. However, current systems remain static: they are trained offline, operate with fixed perceptual assumptions, and lack mechanisms to handle ambiguous instructions like changing environments or novel tasks. To deploy robots reliably in daily scenarios, they must be able to learn continuously from experience, reduce uncertainty through active perception, and acquire new skills from interaction. My research interest is therefore in active learning and sequential decision-making under uncertainty for long-term robotic manipulation, especially in high-dimensional visual environments.

2 Motivation and Problem Setting

There are three limitations that drive my motivations. First, robots encounter unknown tasks when receiving natural language prompts but currently lack principled ways to detect task novelty or request examples. Second, manipulation often fails due to partial observability: occlusions and clutter make grasp success uncertain. Many existing VLA policies operate only on the current camera frame and passively accept uncertainty. Third, human instructions are frequently ambiguous or poorly grounded in the visual scene, yet robots seldom ask for clarifying questions. These challenges will become critical as domestic and industrial assistance robots operate over a long time.

Current research tackles each issue only partially. Active view selection methods [5, 8, 9] typically optimize information gain for reconstruction, but do not quantify how views improve manipulation success. Lifelong learning techniques [1, 3, 2] enable robots acquiring new skills, but how to handle its memory or skills fusion still under exploration. Recent robotic dialog systems [4, 7, 6] can ask clarifying questions but often trigger heuristically rather than from calibrated uncertainty. I believe a unifying perspective is possible: all three problems arise from uncertainty-driven sequential decision-making, where the system actively selects perceptual, interactive, or learning actions to maximize expected improvement.

3 Research Aim and Originality

I propose to develop a Lifelong Active Learning Framework for Robotic Manipulation, grounded in information-theoretic principles. Unlike reconstruction-centric active perception, I will extend some information-theoretic metrics like Fisher Information to quantify how valuable new observations are for manipulation quality, not only scene accuracy. In addition, novelty-aware language grounding can trigger active queries when the robot detects unfamiliar skill requirements, and ambiguous instructions can initiate clarification questions. Together, these enable a robot to learn continuously and robustly through closed-loop perception and interaction.

4 Research Directions

4.1 Active Perception for Manipulation under Occlusion

I will extend Fisher Information-based criteria to measure how informative viewpoint changes are for task success. By calculating the Fisher information about the scene and grasping for the candidate view, the robot will move its camera to reduce uncertainty. This bridges grasp-success-driven viewpoint selection with analytical information methods, building upon my previous work on Fisher Information-guided view selection for scene representation.

4.2 Active Clarification through Uncertainty-Grounded Dialogue

Using calibrated multi-modal uncertainty in grounding like entropy over referred objects, the robot can detect ambiguous language. If there is a mismatch between scene and prompts, instead of moving aimlessly, it will ask for detailed clarifications. Reinforcement learning can optimize the trade-off between asking questions and executing uncertain actions. This component contributes robustness and safety in human–robot interaction.

4.3 Detection and Acquisition of Novel Skills

I will model the distribution over known task embeddings. When a language instruction lies outside the support of existing skills, the robot will request demonstrations or corrective feedback. Parameter-efficient adaptation will enable the integration of new skills without catastrophic forgetting. This supports long-term deployment without offline retraining. 4. Unified Lifelong Active Learning Architecture Finally, I will integrate the above into a multi-timescale controller. A high-level decision module chooses whether to:

- take an informative observation,
- ask a clarifying question,
- request a demonstration, or
- execute the current plan.

Action selection will maximize expected utility and minimize task uncertainty. This unified formulation is novel: while prior work explores each behavior separately, no existing system coordinates them under a single uncertainty-driven policy.

5 Preliminary Knowledge and Skills

My background aligns strongly with this agenda. I have worked extensively on active learning, uncertainty modeling, and sequential decision-making in high-dimensional visual environments. A first-author submission to 3DV develops Fisher Information-based next-best-view selection for semantic Gaussian Splatting, demonstrating my ability to implement principled uncertainty metrics. My accepted works in IROS and IEEE Transactions on Automation Science and Engineering address ambiguity in semantic relocalization and occlusion-aware multi-object tracking, respectively, reinforcing my experience with temporal uncertainty. I also have experience in robot trajectory generation and control, manipulation with RGB-D observations, and integrating action representations into generative video models. These skills will help bridge the gap between theoretical active learning and embodied systems.

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